

Nicholas A. Hernandez

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science — Computational Perception and Robotics

Atlanta, GA

Fall 2026 - Expected 2028

Boston College, Morrissey College of Arts and Sciences

Bachelor of Science in Computer Science

Chestnut Hill, MA

Fall 2022 - Spring 2026

- **GPA: 3.91 Summa Cum Laude** | Phi Beta Kappa, Alpha Sigma Nu, Dean's List
- **Relevant Coursework:** Machine Learning, Robotics, NLP, Algorithms, Linear Algebra, Logic & Computation, Computer Systems, Computer Organization, Multivariable Calculus, Randomness & Computation, GIS Spatial Data.
- **Groups:** Computer Science Society, Machine Intelligence Group, Emerging Leaders Program, Cuban American Student Association

SKILLS

Programming Languages: Python, Java.

ML & Robotics: PyTorch, Hugging Face, scikit-learn, NumPy, Pandas, ROS2, OpenCV

Tools: Git/GitHub, Linux/Unix, Jupyter

RESEARCH EXPERIENCE

REU at Boston College

Natural Language Processing (NLP) Researcher

Boston, MA

May 2025 - June 2026

- Built and preprocessed multilingual speech datasets (10+ hours per language) across 13 languages
- Trained and fine-tuned Wav2Vec2/XLSR-53 models using PyTorch and Hugging Face; developed preprocessing and vocabulary-unification pipelines to support multilingual fine-tuning
- Applied cross-linguistic similarity metrics (e.g., ATDS) to guide transfer learning, reducing Word Error Rate (WER) by up to 68%, and improving ASR model performance

NSF REU at the University of Houston

Machine Learning and Artificial Intelligence Researcher

Houston, TX

May 2024 - August 2024

- Supported multilingual toxicity-detection experiments using XLM-RoBERTa, investigating cross-lingual transfer across Romance languages
- Built a Romance-language toxicity dataset by sourcing, cleaning, and combining public datasets, producing 85,000+ examples for binary classification

PROJECTS

Senior Honors Thesis: Cross-Linguistic Transfer Learning for Machine Translation

Fall 2025 - Spring 2026

- Fine-tuned NLLB-200-distilled-600M across 14 Formosan languages using FormosanBank parallel corpora; designed and ran four experimental pipelines (single- vs. two-stage fine-tuning, supplementary-language combinations, cross-family FLORES transfer)
- Found two-stage fine-tuning with triple supplementary languages (Amis, Atayal, Rukai) most effective, and baseline BLEU strongly correlated with corpus size ($r = 0.827$); results submitted as part of a short paper (ACL Rolling Review)

Robot Car Soccer System

Spring 2026

- Built a real-time OpenCV vision pipeline (HSV masking, contour/circularity analysis) to track a 2-player Bluetooth-controlled robot car match via overhead camera, detecting ball position and goals
- Integrated the Claude API to generate live play-by-play commentary and a Pygame scoreboard UI to track the game

Spatial Analysis of Datacenter Density & Utility Costs

Fall 2025

- Built an ArcGIS spatial analysis pipeline mapping the relationship between datacenter density and residential utility bill costs across Virginia using public infrastructure and utility datasets

MicroRTS Hybrid Reinforcement Learning Agent

Spring 2025

- Built a hybrid agent combining tree-search planning with a PPO-based control model for the MicroRTS real-time strategy environment, balancing economic development against relative military strength
- Achieved a TrueSkill rating of 33.7, outperforming the CoacAI baseline agent (33.4)

VOLUNTEER & LEADERSHIP EXPERIENCE

John M Barry Boys and Girls Club STEM Center

Chestnut Hill, MA

- Taught 10 hrs/week; designed daily exercises and gave Spanish-language instruction for non-English speakers

President of the Cuban American Student Association

Chestnut Hill, MA

- Organize 10–15 culturally enriching events per semester for 400+ members with the Office of Student Involvement and campus clubs